Joshua Shinkle

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INTERESTS

My research interests are in software/systems security and privacy. My research leverages program analysis, reverse engineering, and human-centered methods to study vulnerabilities in mobile phones across both the hardware and software layers.

EDUCATION

- **Ph.D. in Computer Science advised by Antonio Bianchi**
  - Purdue University, *August '24 - May '29 (expected)*
- **B.S. in Computer Science - Cybersecurity**
  - Taylor University, *August '20 - May '24, Summa Cum Laude*

EXPERIENCE

Purdue University - Graduate Research Assistant - PurSec Lab *August '24 - Present*
- Developed a contention-based timing side-channel for mobile NPUs
- Investigated the privacy risks of military-marketed mobile apps
- Produced Android userspace exploits for DARPA's Intelligent Generation of Tools for Security (INGOTS) program

Peraton Labs - Cybersecurity Research Internship *Summer '26*
- Implemented novel sensor fusion capabilities on Android phones by leveraging Bayesian inference networks

Peraton Labs - Cybersecurity Research Internship *Summer '25*
- Built an Android utility that leverages kernel tracing and interprocess communication introspection to trace messages between apps, telephony, Radio Interface Layer (RIL), and baseband firmware

Peraton Labs - Cybersecurity Research Internship *Summer '24*
- Trained a multitask deep learning recommender system to predict mobile app usage based on user demographics
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Peraton Labs - Cybersecurity Research Internship *Summer '23*
- Built a python tool to schedule/generate Android automations to mimic human activity
- Worked as part of a team to repurpose mobile phone communication channels to enable secure communications

Lockheed Martin ATL - Cybersecurity Research Internship *Summer '22*
- Reverse engineered Windows applications
- Windows kernel debugging

Anchorage Digital - Software Engineering Internship *Summer ‘21*

- Worked as part of a team to prototype a new feature critical to supporting DeFi apps on the Ethereum blockchain